

AP CALCULUS AB - UNIT 2

Guided Notes 2.3: The Derivative: Graphs, Differentiability, and Continuity



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we estimate a derivative from a graph, numerical table, or local data?

- Estimate a derivative from a graph, numerical table, or local data.
- Compare forward, backward, and centered differences.
- Judge the reliability of an estimate using interval size and local smoothness.
- Explain why differentiability at a point implies continuity there.
- Identify discontinuities, corners, cusps, and vertical tangents as sources of nondifferentiability.
- Determine parameter values that make a piecewise function continuous and differentiable.

Key Knowledge, Vocabulary, and Notation

Derivative Estimation:
Forward difference

$$\frac{f(a+h) - f(a)}{h}$$

Derivative Estimation:
Backward difference

$$\frac{f(a) - f(a-h)}{h}$$

Derivative Estimation:
Centered difference

$$\frac{f(a+h) - f(a-h)}{2h}, \text{ often more accurate for smooth functions.}$$

Derivative Estimation:
Graphical estimate

Approximate the slope of a carefully chosen tangent line, then use rise over run.

Differentiability and Continuity: Necessary condition

If $f'(a)$ exists, then f is continuous at a .

Differentiability and Continuity: Not sufficient

Continuity alone does not guarantee differentiability; $|x|$ is continuous but has a corner at 0.

Differentiability and Continuity: One-sided derivatives

$f'(a)$ exists only when the left and right derivative limits exist finitely and are equal.

Differentiability and Continuity: Common failures

Discontinuity, corner, cusp, vertical tangent, or sufficiently wild oscillation.

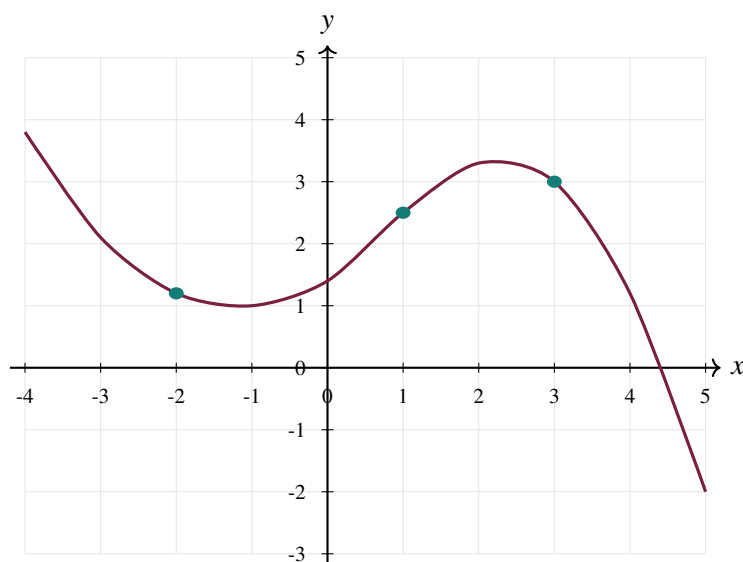
Concept Development and Strategy

1. **Derivative Estimation.** Use points close to the target, but not so close that rounding error dominates. When a table is rounded, several nearby estimates should be compared.
2. **Derivative Estimation.** A centered difference balances information from both sides and cancels some first-order error for smooth functions.
3. **Derivative Estimation.** At a corner or cusp, left and right slope estimates approach different values or become unbounded; reporting one centered number can hide nondifferentiability.
4. **Differentiability and Continuity.** To test a piecewise join, first impose continuity. Only after the function values match should you impose equality of the one-sided derivatives.
5. **Differentiability and Continuity.** A vertical tangent may have a visually well-defined tangent line but its slope is not a finite real number, so the ordinary derivative does not exist.
6. **Differentiability and Continuity.** When justifying, name the exact condition that fails: unequal one-sided slopes, unbounded slope, or lack of continuity.

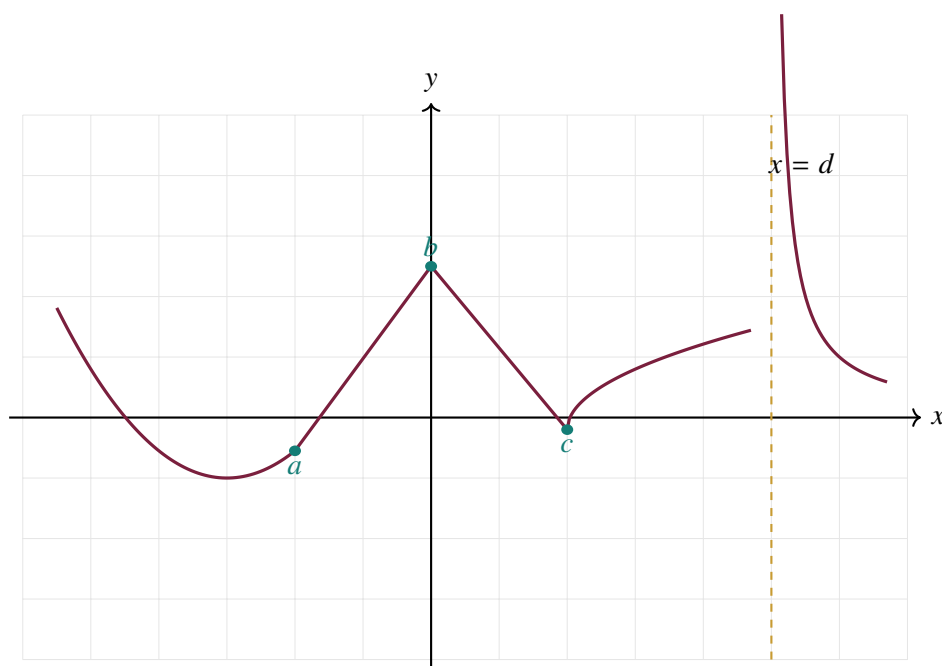
Shared Representation / Data

Derivative Estimation

x	1.80	1.90	1.99	2.00	2.01	2.10	2.20
$f(x)$	4.812	4.947	5.085	5.100	5.115	5.247	5.388



Differentiability and Continuity

**Worked Example: Derivative Estimation - Centered estimate**

Given $f(1.9) = 4.61$ and $f(2.1) = 5.43$, estimate $f'(2)$.

Answer and Reasoning

$$\frac{5.43 - 4.61}{0.2} = 4.1.$$

Worked Example: Derivative Estimation - From a graph

A tangent line through the point of interest also passes approximately through $(1, 2.2)$ and $(5, 7.0)$. Estimate the derivative.

Answer and Reasoning

$$\frac{7.0 - 2.2}{5 - 1} = 1.2.$$

Worked Example: Derivative Estimation - Detecting a corner

Near $x = 0$, left secant slopes approach -2 while right secant slopes approach 3 . What is $f'(0)$?

Answer and Reasoning

It does not exist because the one-sided derivative limits disagree.

Worked Example: Differentiability and Continuity - Continuous but not differentiable

Analyze $f(x) = |x - 2|$ at $x = 2$.

Answer and Reasoning

The function is continuous. The left derivative is -1 and the right derivative is 1 , so $f'(2)$ does not exist.

Worked Example: Differentiability and Continuity - Piecewise differentiability

Find m, b so $f(x) = mx + b$ for $x < 1$ and $f(x) = x^2 + 1$ for $x \geq 1$ is differentiable at 1.

Answer and Reasoning

Continuity gives $m + b = 2$; slope matching gives $m = 2$. Hence $b = 0$.

Worked Example: Differentiability and Continuity - Vertical tangent

For $f(x) = \sqrt[3]{x}$, discuss differentiability at 0.

Answer and Reasoning

The difference quotient is $h^{-2/3} \rightarrow +\infty$, so there is a vertical tangent and no finite derivative at 0.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Derivative Estimation: Use $f(2.9) = 8.41$ and $f(3.1) = 9.61$ to estimate $f'(3)$ by a centered difference.

2. Derivative Estimation: Use $g(4) = 11.2$ and $g(4.05) = 11.38$ to make a forward estimate of $g'(4)$.

3. Derivative Estimation: Using the shared table, estimate $f'(2)$ with $h = 0.01$ and with $h = 0.10$.

4. Derivative Estimation: From the shared graph, rank the derivative estimates at $x = -2, 1, 3$ from least to greatest.

5. Derivative Estimation: For a smooth function, forward estimates with $h = .2, .1, .05$ are 3.40, 3.20, 3.10. Predict the derivative and explain the pattern.

6. Derivative Estimation: Use $f(a - 2h) = 4.7$, $f(a - h) = 4.9$, $f(a + h) = 5.5$, $f(a + 2h) = 6.0$ to compare centered estimates with radii h and $2h$.

7. Derivative Estimation: Show algebraically that the centered difference for $f(x) = x^3$ at a equals $3a^2 + h^2$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Derivative Estimation: Which estimate generally uses data from both sides of a ?

- (A) forward difference (B) backward difference
(C) centered difference (D) average function value

2. Derivative Estimation: If left derivative estimates approach 4 and right derivative estimates approach 4, the best conclusion is

- (A) $f(a) = 4$ (B) $f'(a)$ is likely 4 if the function is defined appropriately
(C) the function has a corner (D) the derivative is 8

3. Derivative Estimation: Using the shared table, the centered estimate of $f'(2)$ with $h = .1$ is

- (A) 0.75 (B) 1.5
(C) 2.7 (D) 5.1

Common Errors and AP Precision

- State the derivative notation, input value, sign, and units when interpreting a rate in context.
- Check differentiability before using a tangent slope, and keep the order of Product and Quotient Rule terms accurate.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Differentiability and Continuity: Prove differentiability implies continuity using $f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a)$.

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